

The clock signals are used in sequential logic circuits to _____?

- ☐ Tell how much time has elapsed since the system was turned on
- ☒ Tell when to write to memory
- ☐ Carry parallel data signals
- ☐ Tell the time of the day

Popular application of flip-flop are _____?

☒ Cache/Registers

☐ Multiplexers

☐ Decoders

☐ None

On a leading edge-triggered S-R flip-flop, the outputs reflect the input condition when

- ☒ The clock pulse transitions from LOW to HIGH
- ☐ The clock pulse is HIGH
- ☐ The clock pulse transitions from HIGH to LOW
- ☐ The clock pulse is LOW

The flip-flops which have not any invalid states are _____?

☐ S-R, J-K, T

☐ S-R, J-K, D

☒ J-K, D, T

☐ J-K, D, S-R

The main difference between JK and RS flip-flop is that

- ☐ JK flip-flop has a Clear-Preset inputs
- ☐ There is a feedback in JK lip-lop
- ☐ JK flip flop needs a clock pulse
- ☒ JK flip-flop accepts both inputs as 1

A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates?

- ☐ XOR or XNOR gates
- ☐ AND or NOR gates
- ☒ NAND or NOR gates
- ☐ None

The characteristic equation of S-R latch is _____

☐ $Q^* = SR + QR$

☐ $Q^* = S'R + QR$

☐ $Q^* = S'R + Q'R$

☒ $Q^* = (S + Q)R'$

The PRE and CLR inputs of a JK flip-flop set and reset the Q output

- ☐ Only if J and K are not connected to anything.
- ☐ Only after the clock input goes high.
- ☒ Unconditionally.
- ☐ Only if $J = 0$ and $K = 0$.

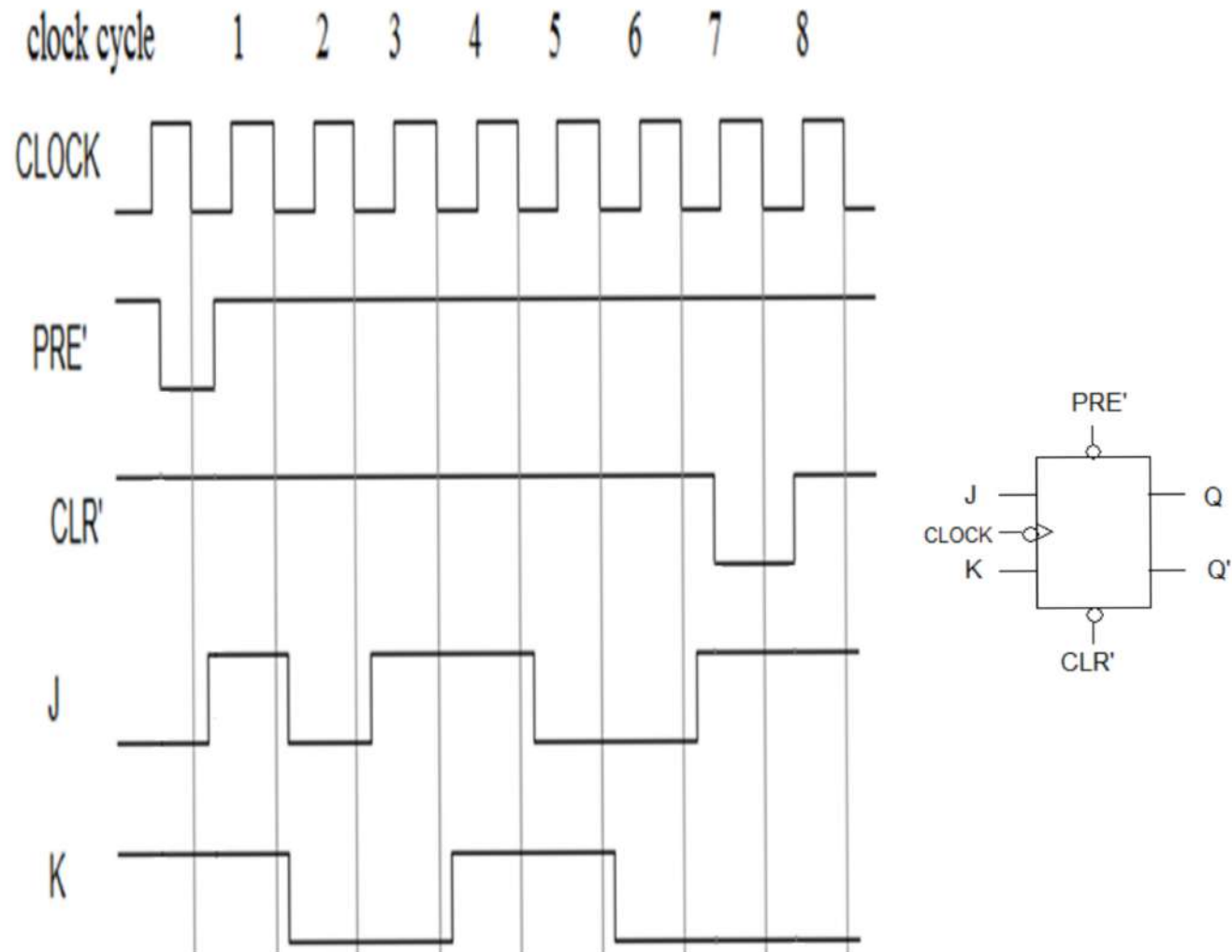
Which of the following is correct for a gated D-type flip-flop?

- ☒ The Q output is either SET or RESET as soon as the D input goes HIGH or LOW
- ☐ Only one of the inputs can be HIGH at a time
- ☐ The output toggles if one of the inputs is held HIGH
- ☐ The output complement follows the input when enabled

For the JK flip flop and if the values of CLR' and PRE' are as shown in the figure.

Complete the timing diagram. Then answer the question of

What is the value of Q in clock cycles (1, 3, and 5)?



☐ None

☒ 1 0 0

☐ 1 0 1